

Program 4

STEP 1: Create Project Folder

STEP 2: Create `package.json`

STEP 3: Install Express Dependency

STEP 4: Create Application Code (`index.js`)

STEP 5: Create Dockerfile

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ mkdir 4rthprg
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ cd 4rthprg/
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ npm init -y
Wrote to /home/1rv24mc077_prateek_bhandari/dockerprgs/4rthprg/package.json:

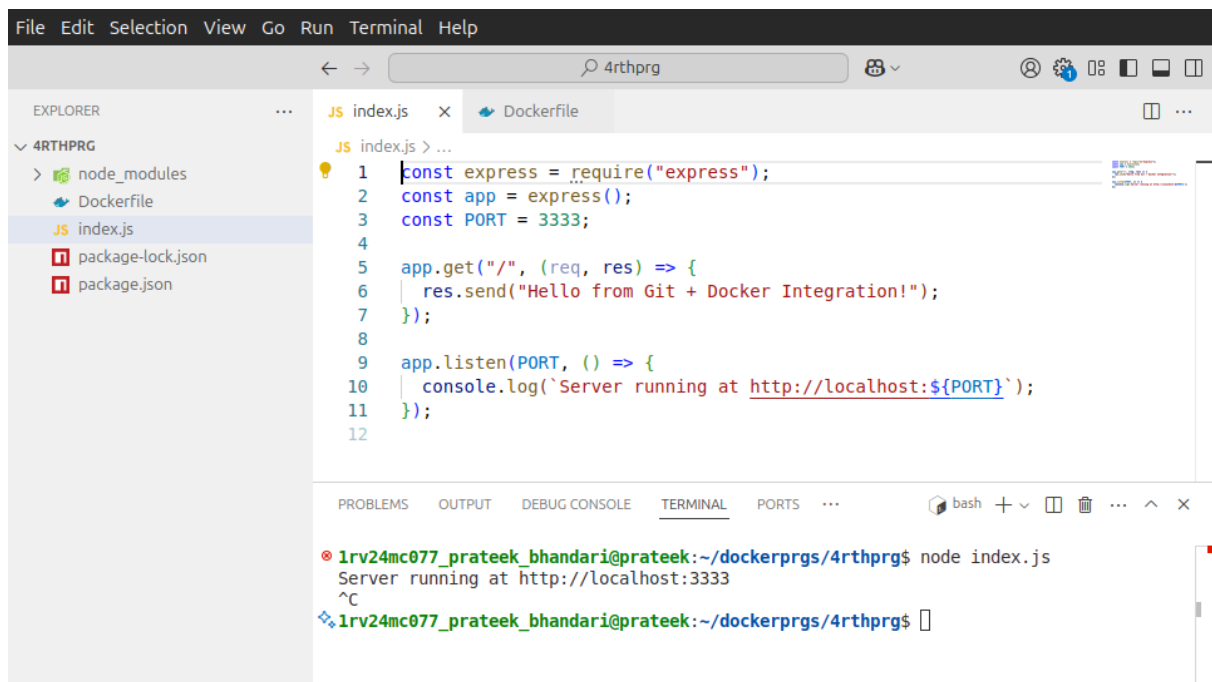
{
  "name": "4rthprg",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ npm install express

added 65 packages, and audited 66 packages in 4s

22 packages are looking for funding
  run `npm fund` for details

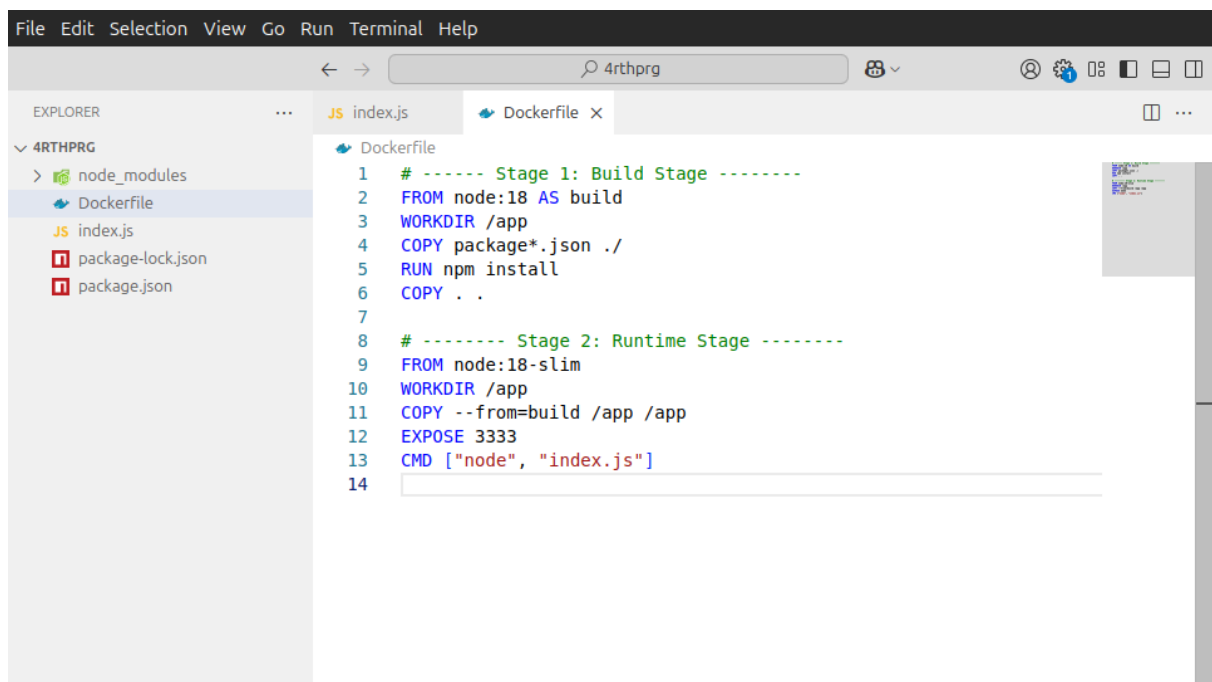
found 0 vulnerabilities
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ nano index.js
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ touch Dockerfile
```



The screenshot shows the VS Code editor with the Explorer sidebar on the left displaying the file structure of a project named '4RTHPRG'. The files listed are 'node_modules', 'Dockerfile', 'index.js', 'package-lock.json', and 'package.json'. The 'index.js' file is open in the editor, showing the following code:

```
1 const express = require("express");
2 const app = express();
3 const PORT = 3333;
4
5 app.get("/", (req, res) => {
6   res.send("Hello from Git + Docker Integration!");
7 });
8
9 app.listen(PORT, () => {
10   console.log(`Server running at http://localhost:${PORT}`);
11 });
12
```

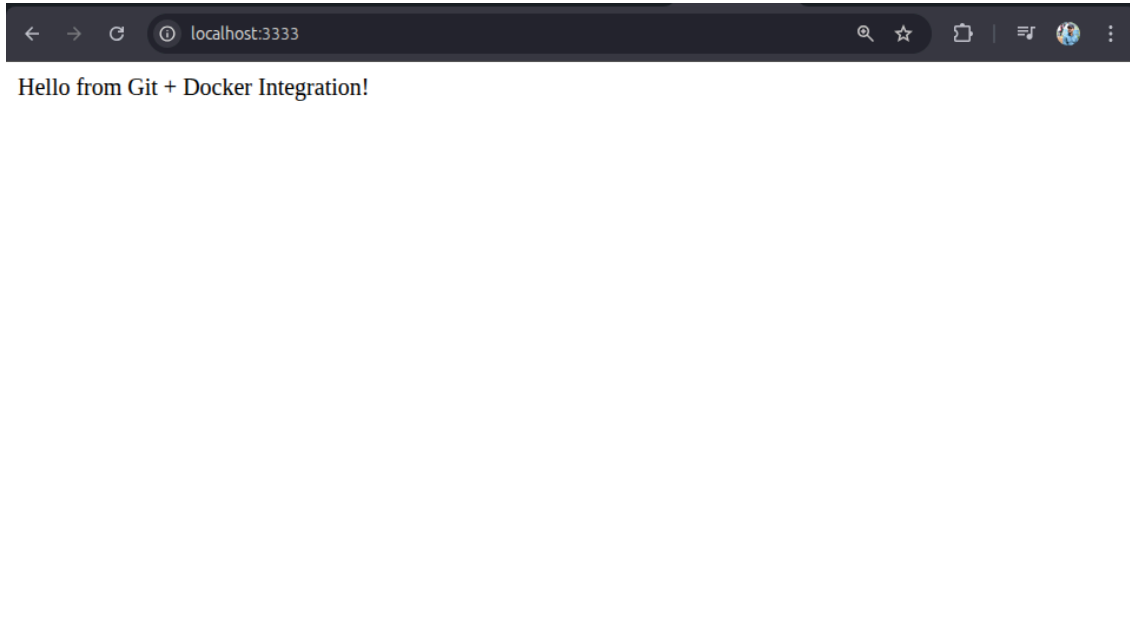
The terminal at the bottom shows the command 'node index.js' being executed, resulting in the output 'Server running at http://localhost:3333'. The terminal prompt is '1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg\$'.



The screenshot shows the VS Code editor with the Explorer sidebar on the left displaying the file structure of a project named '4RTHPRG'. The files listed are 'node_modules', 'Dockerfile', 'index.js', 'package-lock.json', and 'package.json'. The 'Dockerfile' file is open in the editor, showing the following content:

```
1 # ----- Stage 1: Build Stage -----
2 FROM node:18 AS build
3 WORKDIR /app
4 COPY package*.json ./
5 RUN npm install
6 COPY . .
7
8 # ----- Stage 2: Runtime Stage -----
9 FROM node:18-slim
10 WORKDIR /app
11 COPY --from=build /app /app
12 EXPOSE 3333
13 CMD ["node", "index.js"]
14
```

STEP 6: Test App Locally



STEP 7: Build Docker Image

```
1rv24nc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ docker build -t git-docker .
[+] Building 10.3s (15/15) FINISHED
=> [internal] load build definition from Dockerfile                                0.0s
=> => transferring dockerfile: 308B                                              0.0s
=> [internal] load metadata for docker.io/library/node:18                      3.8s
=> [internal] load metadata for docker.io/library/node:18-slim                 3.8s
=> [auth] library/node:pull token for registry-1.docker.io                     0.0s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 2B                                                    0.0s
=> [build 1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d2 0.0s
=> [internal] load build context                                                0.2s
=> => transferring context: 2.16MB                                              0.2s
=> [stage-1 1/3] FROM docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e 6.1s
=> => resolve docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e 0.0s
=> => sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651 6.49kB / 6.49kB 0.0s
=> => sha256:fc3faf127a182135fd956e68d570b1932a758f8008866d8dd6e131cf8 1.93kB / 1.93kB 0.0s
=> => sha256:101e0128c8ea90af6e5eba2abbae8486503c6383c35cb30e2c60842a5 6.54kB / 6.54kB 0.0s
=> => sha256:61320b01ae5e0798393ef25f2dc72faf43703e60ba089b07d7170ac 28.23MB / 28.23MB 2.9s
=> => sha256:b98d3ae1ab80d768fc7be41c07c4757d205cc33d2d26261b8273defb4 3.31kB / 3.31kB 1.1s
=> => sha256:b1831021e35a69864dfc4c89eab9ab1232cf3508c787b082c55ee38 38.25MB / 38.25MB 3.5s
=> => sha256:c768ab8cba73bb84835e67ed974c93e3e0e3ce1f67a73802422d66021 1.71MB / 1.71MB 2.3s
=> => sha256:8c994cf49dd19aec88c96926e1a13b36a8f3e159942a5a2d0a9e2dc3e9e6c 448B / 448B 3.0s
=> => extracting sha256:61320b01ae5e0798393ef25f2dc72faf43703e60ba089b07d7170acbabbf8f 1.5s
=> => extracting sha256:b98d3ae1ab80d768fc7be41c07c4757d205cc33d2d26261b8273defb454553 0.0s
=> => extracting sha256:b1831021e35a69864dfc4c89eab9ab1232cf3508c787b082c55ee386c5c852 1.4s
=> => extracting sha256:c768ab8cba73bb84835e67ed974c93e3e0e3ce1f67a73802422d66021fc9a0 0.0s
=> => extracting sha256:8c994cf49dd19aec88c96926e1a13b36a8f3e159942a5a2d0a9e2dc3e9e6c3 0.0s
=> CACHED [build 2/5] WORKDIR /app                                             0.0s
=> [build 3/5] COPY package*.json ./                                           0.1s
=> [build 4/5] RUN npm install                                                  2.7s
=> [build 5/5] COPY . .                                                         0.4s
=> [stage-1 2/3] WORKDIR /app                                                  0.1s
=> [stage-1 3/3] COPY --from=build /app /app                                   0.1s
=> exporting to image                                                         0.1s
=> => exporting layers                                                         0.1s
=> => writing image sha256:aa8b38ddce31952af880406ea0d6e89250119e4a0fd483bd80d93b5e494 0.0s
=> => naming to docker.io/library/git-docker                                  0.0s
1rv24nc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
git-docker:latest	aa8b38ddce31	194MB	0B	
kicbase/stable:v0.0.48	c6b5532e987b	1.31GB	0B	

STEP 8: Run Docker Container

```
=> => exporting layers 0.1s
=> => writing image sha256:aa8b38ddce31952af880406ea0d6e89250119e4a0fd483bd80d93b5e494 0.0s
=> => naming to docker.io/library/git-docker 0.0s
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
git-docker:latest	aa8b38ddce31	194MB	0B	
kicbase/stable:v0.0.48	c6b5532e987b	1.31GB	0B	U
mongo:latest	08b59ba09096	906MB	0B	U

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ docker run -it --name application4 -p 3333:3333 git-docker:latest
Server running at http://localhost:3333
```

STEP 9: Create GitHub Repository

STEP 10: Generate GitHub Personal Access Token (PAT)

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#)
Required fields are marked with an asterisk ().*

1

General

Owner *

PrateekBhandari07

Repository name *

PRG4

PRG4 is available.

Great repository names are short and memorable. How about [automatic-potato?](#)

Description

0 / 350 characters

2

Configuration

Choose visibility *

Public

Choose who can see and commit to this repository

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

STEP 11: Initialize Git Locally

```
^C1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1rv24mc077_prateek_bhandari/dockerprgs/4rthprg/.git/
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ git add .
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ git commit -m "committing all files"
[master (root-commit) 441b7d7] committing all files
 589 files changed, 61914 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 index.js
 create mode 100644 node_modules/.package-lock.json
 create mode 100644 node_modules/accepts/HISTORY.md
 create mode 100644 node_modules/accepts/LICENSE
 create mode 100644 node_modules/accepts/README.md
 create mode 100644 node_modules/accepts/index.js
 create mode 100644 node_modules/accepts/package.json
 create mode 100644 node_modules/body-parser/LICENSE
 create mode 100644 node_modules/body-parser/README.md
 create mode 100644 node_modules/body-parser/index.js
 create mode 100644 node_modules/body-parser/lib/read.js
 create mode 100644 node_modules/body-parser/lib/types/json.js
 create mode 100644 node_modules/body-parser/lib/types/raw.js
```

STEP 12: Link Remote Repository

```
 create mode 100644 node_modules/wrappy/README.md
 create mode 100644 node_modules/wrappy/package.json
 create mode 100644 node_modules/wrappy/wrappy.js
 create mode 100644 package-lock.json
 create mode 100644 package.json
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ git branch -M main
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ git remote add origin https://github.com/PrateekBhandari07/PRG4.git
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$ git push -u origin main
Enumerating objects: 678, done.
Counting objects: 100% (678/678), done.
Delta compression using up to 4 threads
Compressing objects: 100% (627/627), done.
Writing objects: 100% (678/678), 668.21 KiB | 4.91 MiB/s, done.
Total 678 (delta 122), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (122/122), done.
To https://github.com/PrateekBhandari07/PRG4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/4rthprg$
```

STEP 13: Push Code to GitHub

The screenshot shows a GitHub repository page for a user named 'PrateekBhandari07' with the repository name 'PRG4'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The repository contains a commit from 3 minutes ago with the message 'committing all ...' and a commit hash of '441b7d7'. The commit includes files: 'node_modules', 'Dockerfile', 'index.js', 'package-lock.json', and 'package.json'. The repository also has a README section with a button to 'Add a README'. The right sidebar shows the 'About' section with a note that no description, website, or topics are provided. It also shows sections for 'Releases' (no releases published), 'Packages' (no packages published), and 'Languages' (Dockerfile 51.3%, JavaScript 48.7%).

PrateekBhandari07 / PRG4

Code Issues Pull requests Actions Projects Wiki

PRG4 Public Pin Watch 0 Fork 0 Star 0

m... Go to file + <> Code About

PrateekBhandari07 commi... 441b7d7 · 3 minutes ago

node_modules	committing all ...	3 minutes ago
Dockerfile	committing all ...	3 minutes ago
index.js	committing all ...	3 minutes ago
package-lock.json	committing all ...	3 minutes ago
package.json	committing all ...	3 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages

Dockerfile	51.3%
JavaScript	48.7%